

$$1. \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 & 2 & 0 & -1 & 0 \\ 2 & 1 & 0 & 0 & -1 & 0 & 0 & 1 \\ 2 & 0 & 1 & 1 & 0 & 4 & 0 & 0 \end{array} \right] \begin{array}{l} 0 \\ 1 \\ 0 \\ 0 \end{array}$$

$$R_3 = R_3 - 2R_1, \quad R_4 = R_4 - 2R_1, \quad R_2 - R_1 = R_2$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} 1 \\ 0 \\ 0 \\ 0 \end{array}$$

$$R_2 = \frac{R_2}{2}, \quad R_3 = R_3 - R_2$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1/2 & 1 & 0 & 1/2 & 0 \\ 0 & 0 & 1 & -1/2 & -2 & -1/2 & 0 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} 0 \\ 0 \\ 0 \\ 0 \end{array}$$

$$R_3 = -2R_3$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1/2 & 1 & 0 & 1/2 & 0 \\ 0 & 0 & 1 & -4 & 4 & 1 & -2 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} 0 \\ 0 \\ 0 \\ 0 \end{array}$$

$$R_2 = R_2 - \frac{R_3}{2}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 4 & 4 & 0 & -2 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 & 0 & 1 \end{array} \right]$$

$$R_4 = R_4 - R_3$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 4 & 4 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 & -6 & -1 & 2 & 1 \end{array} \right]$$

$$R_4 = -R_4 / 2$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 4 & 4 & 0 & -2 & 0 \\ 0 & 0 & 0 & -1 & 3 & 1/2 & -1 & -1/2 \end{array} \right]$$

$$R_1 = R_1 - R_4$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -2 & -1/2 & 1 & 1/2 \\ 0 & 1 & 0 & -1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 4 & 4 & 0 & -2 & 0 \\ 0 & 0 & 0 & -1 & 3 & 1/2 & -1 & -1/2 \end{array} \right]$$

$$R_2 = R_2 + R_4$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -2 & -1/2 & 1 & 1/2 \\ 0 & 1 & 0 & 0 & 1 & 1/2 & 0 & -1/2 \\ 0 & 0 & 1 & 0 & 4 & 1/2 & -2 & 0 \\ 0 & 0 & 0 & 1 & 3 & 1/2 & -1 & -1/2 \end{array} \right] = A \cdot b$$

$$R_3 = R_3 - 4R_4$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -2 & -1/2 & 1 & 1/2 \\ 0 & 1 & 0 & 0 & 1 & 1/2 & 0 & -1/2 \\ 0 & 0 & 1 & 0 & -8 & -1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 3 & 1/2 & -1 & -1/2 \end{array} \right] = R$$

Answer

$$\begin{bmatrix} -2 & -1/2 & 1 & 1/2 \\ 1 & 1/2 & 0 & -1/2 \\ -8 & -1 & 2 & 2 \\ 3 & 1/2 & -1 & -1/2 \end{bmatrix} \cdot A$$

$$\left[\begin{array}{ccc|ccc} 0 & 0 & 1/2 & 0 & 1/2 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1/2 & 1 & 1/2 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 0 & 0 & 1/2 & 0 & 1/2 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1/2 & 1 & 1/2 & 0 \end{array} \right]$$

$$2. a. A = \begin{bmatrix} 2 & 1 & 0 \\ -2 & 3 & 7 \\ 3 & 0 & 4 \end{bmatrix} \quad \begin{array}{ccc|ccc} 2 & 1 & 0 & 1 & 0 & 0 \\ -2 & 3 & 7 & 0 & 1 & 0 \\ 3 & 0 & 4 & 0 & 0 & 1 \end{array}$$

$$\begin{bmatrix} 2 & 1 & 0 & 1 & 0 & 0 \\ -2 & 3 & 7 & 0 & 1 & 0 \\ 3 & 0 & 4 & 0 & 0 & 1 \end{bmatrix} \quad R_2 = R_2 + R_1$$

$$R_1 = R_1 / 2$$

$$\begin{bmatrix} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ -2 & 3 & 7 & 0 & 1 & 0 \\ 3 & 0 & 4 & 0 & 0 & 1 \end{bmatrix} \quad \text{Answer}$$

$$R_2 = R_2 + 2R_1$$

$$\begin{bmatrix} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 4 & 7 & 1 & 1 & 0 \\ 3 & 0 & 4 & 0 & 0 & 1 \end{bmatrix}$$

$$R_3 = R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 4 & 7 & 1 & 1 & 0 \\ 0 & -3/2 & 4 & -3/2 & 0 & 1 \end{bmatrix}$$

$$R_2 = R_2 / 4$$

$$\begin{bmatrix} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 7/4 & 1/4 & 1/4 & 0 \\ 0 & -3/2 & 4 & -3/2 & 0 & 1 \end{bmatrix}$$

$$R_1 = R_1 - \frac{R_2}{2}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -7/8 & 3/8 & -1/8 & 0 \\ 0 & 1 & 7/4 & 1/4 & 1/4 & 0 \\ 0 & -3/2 & 5/4 & -3/2 & 0 & 1 \end{array} \right]$$

$$R_3 = R_3 + \frac{3R_2}{2}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -7/8 & 3/8 & -1/8 & 0 \\ 0 & 1 & 7/4 & 1/4 & 1/4 & 0 \\ 0 & 0 & 53/8 & -9/8 & 3/8 & 1 \end{array} \right]$$

$$R_3 = \frac{8R_3}{53}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -7/8 & 3/8 & -1/8 & 0 \\ 0 & 1 & 7/4 & 1/4 & 1/4 & 0 \\ 0 & 0 & 1 & -9/53 & 3/53 & 8/53 \end{array} \right]$$

$$R_1 = R_1 + \frac{7R_3}{8}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 12/53 & -4/53 & 7/53 \\ 0 & 1 & 7/4 & 1/4 & 1/4 & 0 \\ 0 & 0 & 1 & -9/53 & 3/53 & 8/53 \end{array} \right]$$

$$R_2 = R_2 - \frac{7R_3}{4}$$

$$R_2 = R_2 - \frac{7R_3}{4}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 12/53 & -4/53 & 7/53 \\ 0 & 1 & 0 & 29/53 & 8/53 & -14/53 \\ 0 & 0 & 1 & -9/53 & 3/53 & 8/53 \end{array} \right]$$

Answer

$$R_2 = R_2 + \frac{9R_3}{5}$$

$$\left[\begin{array}{ccc|ccc} 12/53 & -4/53 & 7/53 & 0 & 1 & 0 \\ 29/53 & 8/53 & -14/53 & 0 & 0 & 1 \\ -9/53 & 3/53 & 8/53 & 0 & 0 & 0 \end{array} \right] \cdot A$$

b. $2x + y = 53$
 $-2x + 3y + 7z = -53$
 $3x + 4z = 53$

$$R_2 = R_2 + \frac{9R_3}{5}$$

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 0 & x & y & z \\ -2 & 3 & 7 & 53 & -53 & 53 \end{array} \right]$$

A x B

$$R_2 = R_2 + \frac{9R_3}{5}$$

$$Ax = B$$

$$A^{-1}Ax = A^{-1}B$$

$$Ix = A^{-1}B$$

$$x = A^{-1}B$$

Taking A^{-1} both sides

Taking inverse of A from part a

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12/53 & -4/53 & 7/53 \\ 29/53 & 8/53 & -14/53 \\ -9/53 & 3/53 & 8/53 \end{bmatrix} \begin{bmatrix} 53 \\ -53 \\ 53 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} (12/53) \cdot (53) + (-4/53) \cdot (-53) + (7/53) \cdot (53) \\ (29/53) \cdot (53) + (8/53) \cdot (-53) + (-14/53) \cdot (53) \\ (-9/53) \cdot (53) + (3/53) \cdot (-53) + (8/53) \cdot (53) \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12 + 4 + 7 \\ 29 - 8 - 14 \\ -9 - 3 + 8 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 23 \\ 7 \\ -4 \end{bmatrix}$$

ans

$$3. \begin{bmatrix} -b & c & b \\ b & 1 & c \\ 0 & 0 & -a \end{bmatrix}$$

$$\text{determinant} = (-b)(-a) - (c)(-ab) + b(0-0)$$

$$(ab) + (ac) = ab + ac$$

$$= ab(1+c)$$

$$= ab(1+c)$$

So there is an invertible

$$\begin{bmatrix} -b & c & b & | & 1 & 0 & 0 \\ b & 1 & c & | & 0 & 1 & 0 \\ 0 & 0 & -a & | & 0 & 0 & 1 \end{bmatrix}$$

$$R_1 = -\frac{1}{b} R_1$$

$$\begin{bmatrix} 1 & -c/b & -1 & | & -1/b & 0 & 0 \\ b & 1 & c & | & 0 & 1 & 0 \\ 0 & 0 & -a & | & 0 & 0 & 1 \end{bmatrix}$$

$$R_2 = R_2 - bR_1$$

$$\begin{bmatrix} 1 & -c/b & -1 & | & -1/b & 0 & 0 \\ 0 & c+1 & b+c & | & 1 & 1 & 0 \\ 0 & 0 & -a & | & 0 & 0 & 1 \end{bmatrix}$$

$$R_2 = \frac{R_2}{c+1}$$

$$\left[\begin{array}{ccc|ccc} 1 & -c/b & -1 & -1/b & 0 & 0 \\ 0 & 1 & b+1/c+1 & 1/c+1 & 1/c+1 & 0 \\ 0 & 0 & -a & 0 & 0 & 1 \end{array} \right]$$

$$R_1 = R_1 + \frac{c}{b} R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -b+c/bc+b & -1/bc(c+1) & c/bc(c+1) & 0 \\ 0 & 1 & b+1/c+1 & 1/c+1 & 1/c+1 & 0 \\ 0 & 0 & -a & 0 & 0 & 1 \end{array} \right]$$

$$R_3 = -\frac{1}{a} R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -b+c/bc+b & -1/bc(c+1) & c/bc(c+1) & 0 \\ 0 & 1 & b+1/c+1 & 1/c+1 & 1/c+1 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1/a \end{array} \right]$$

$$R_1 = R_1 - \frac{-b+c}{bc+b} R_3, \quad R_2 = R_2 - \frac{b+1}{c+1} R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1/bc(c+1) & c/bc(c+1) & -b+c/ab(c+1) \\ 0 & 1 & 0 & 1/c+1 & 1/c+1 & b+1/a(c+1) \\ 0 & 0 & 1 & 0 & 0 & -1/a \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} -1/bc(c+1) & c/bc(c+1) & -b+c/ab(c+1) \\ 1/c+1 & 1/c+1 & b+1/a(c+1) \\ 0 & 0 & -1/a \end{array} \right] \text{ ans.}$$